Nico Rosamilia

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Clinical Assistant Professor, *NYU School of Professional Studies Tisch Institute* 2026

Research interests: private & public equity and fixed income modeling, sport finance & economics

Ph.D. in Management Engineering, *Politecnico di Milano School of Management* 2025

Awards

Recipient of the EU grant *National Operational Programme on Research and Innovation* 2021/24

Visiting Researcher

ALBA Graduate Business School, The American College of Greece, Athens 2024
The University of Queensland, Brisbane (Australia) 2023

Teaching

Lecturer, NYU SPS 2021/24

- Sport Finance and Economics (graduate level), Financial Management and Accounting in Sport (undergraduate level); Research Methods (undergraduate level)

Reader and Teaching Assistant

- Politecnico di Milano GSOM: Corporate Finance, Sustainable Finance and ESG 2021/24
- CERGE-EI: Microeconomics 2020
- University of Rijeka: EU and fiscal policy 2018/19

Education

M.A. in Economics, *CERGE-EI* 2019/21

Researcher, University of Rijeka 2018/19

M.Sc. in Management Engineering, Politecnico di Milano 2013/15

B.Sc. in Management Engineering, Università di Napoli Federico II 2010/13

Publications and Submissions

AI-Driven ESG Investing: Robust Portfolios with Dynamic Ensemble Learning (joint work with Ferrari, I., Pariani, E., Marazzina, D.). Accepted for publication in: Annals of Operations Research, 2026

We introduce a Dynamic Ensemble Learning Strategy (DELS) combining adaptive Random Forest classification with volatility-based portfolio recalibration. Applied to Euro STOXX 600 stocks, the strategy improves portfolio robustness, ESG alignment, and risk-adjusted performance, outperforming passive investment benchmarks.

Profit or Planet? Both! ESG Drivers of Efficient Portfolios and the Costs of Disclosure. Status: submitted to Decisions in Economics and Finance Open in August 2026. Available at : *SSRN*

Customizing a Random Forest algorithm, I create efficient sustainable portfolios through best-performing firms selection and identify material ESG indicators in 5 global markets

Returns under the Lens: the Importance of ESG Factors (joint work with Ginestroni, G., Marazzina, D.). Published in: Decisions in Economics and Finance, 2026

Our results show that ESG and financial variables independently improve classification performance comparably, suggesting a complementary role in return forecasting.

What's news with you: Price forecasting with global and ESG sentiment scores (joint work with Leone, F., Marazzina, D.). Published in: Finance Research Open, 2025, 100013

We show that deep learning models (N-HiTS), when trained on sentiment data from news (with sentiment scores computed using RoBERTa), significantly outperform the baseline model in forecasting stock prices

Climate Risk and the Erosion of Brand Capital (joint work with Giudici, G., Konstantios, D.). Status: submitted to Long Range Planning in April 2026

We find that the value of a marginal dollar of cash on brand value is significantly lower in firms with high climate risk exposure compared to those with low exposure

AI-Driven ESG Investing: Robust Portfolios with Dynamic Ensemble Learning (joint work with Ferrari, I., Pariani, E., Marazzina, D.). Status: RR in Decisions in Economics and Finance

We introduce the Dynamic Ensemble Learning Strategy (DELS), ensuring an adaptive decision-making process based on a two-stage consensus mechanism, optimizing risk-adjusted returns while aligning with sustainability objectives.

Working papers

Valuation Practice and Theory: Traditional Models and a Jump-Diffusion Framework

Benchmarking traditional valuation models (DCF, Asset-Based, and Comparable Multiples) on M&A, IPO, and VC datasets with a jump-diffusion real-options framework for illiquid assets.

No Gambling Here! A New Comprehensive Risk Index

I combine financial, macroeconomic, and ESG variables into a synthetic measure representing risk with an unsupervised neural network that overcomes linearity

Climate Shocks and Corporate Resilience: Extreme Weather Events and EU Solidarity Funds for Adaptation and Mitigation (joint work with Qi, R.)

The effect of various extreme weather events on firms' financial performance is comparable in magnitude. The European solidarity funds help in the first aftermath, not necessarily in resilience for future events

Forecasting Stock Returns with Bayesian Econometrics and Machine Learning

Bayesian updating info and Machine Learning for prediction and forecasting in the equity market

Should I Stay or Should I go? Transfers, Trade Gains, and Capital Accumulation in a Fiscal Union

The incentive compatibility of the optimal fiscal union model for countries in the European Monetary Union, comparing the trade gains in the currency union with the transfers in the fiscal union

Fiscal stimulus for EMU countries not meeting SGP criteria (joint work with Cvečić, I., Drezgić, S.)

The effects of the 2010 dry-up in the credit supply for Italian non-financial firms on unemployment

Secondary Market for Private Equity and Venture Capital

Discussing the role of PE and VC secondary markets, the type of assets (trophy or trouble) traded, the information asymmetries, and the pricing challenges

Courses

Macroeconomic Forecasting and Analysis in the Machine Learning Era, Bank of Italy

Advanced Bayesian Econometrics: Bayesian Multivariate Models and Forecasting, Università Ca' Foscari

Panel Data Econometrics: Theory and Applications, Università di Bologna

Striving for Research Quality, Alma Mater Studiorum Bologna

Knowledge Dynamics, Industry Evolution, Economic Development, Université Cote D'Azure

It's Not About Teaching, It's About Learning, Università degli Studi di Bergamo

Experience

Economist (PM and Investment Strategist mandate), *Bulltick* FL, US – 2025

- Portfolio management: managing $500M in the Public Markets Portfolios division; Investment Strategist overseeing $4B
- Portfolio strategies: 3 self-developed cross-asset approaches: (i) TVTP Markov switching forecasting; (ii) *custom ML* optimization with a *sentiment* component; (iii) top-down and bottom-up cross-validation for β-hedged market neutral portfolios
- Alternative assets: Liquidity and cash flows budgeting for 5-year commitment pacing plan with J-curve scenarios, PME/KS-PME analytics. Directional options strategies, delta hedging, and volatility-driven positioning

Investment Manager, *GMA International Services, Inc.* FL, US – 2024/25

- M&A valuation: applied jump-diffusion real-options analysis to value illiquid, high-uncertainty targets, complementing DCF and comparable-company valuation
- Portfolio management: equity portfolio allocation using GDP-lag/vol regimes with β-neutral hedges
- Real estate: managed income-producing residential properties in Italy for *La Promessa*

Researcher, University of Rijeka HR – 2018/19

- Macro modeling: multi-country models for fiscal unions with limited commitment frictions, EU state-contingent transfers, and capital accumulation incentives for low-productivity countries. Academic reference: Drezgić, S.
- Lecturer & EU studies: fiscal stimulus, SGP compliance, and labor effects of credit shocks. Lecturer on EMU macroeconomic policy and sovereign risk-sharing mechanisms. Academic reference: Cvećić, I.

Consultant, *IMI Investment Banking* – Be Shaping the Future IT – 2015/16

- Trading desk data and platform: trading desk analytics, market-data workflow and infrastructure development, and trading platform implementation

Professional Appointments During Full-Time Ph.D.

External Associate, *Intellico* IT – 2021/24

- Research-driven applications: defined ESG material drivers for Italian companies using lasso and ensemble trees on Python. AI-enabled solutions for decision-making in finance and insurance. Article: *AI and ESG*

Researcher, European Regional Development Fund *(NOPRI)* 2021/24

- Quantitative models: *machine learning* for risk measurement and the *EU solidarity fund* effects for extreme weather events

Graduate Student Supervisor, Politecnico di Milano IT – 2021/24

- Mentorship: support students in high-quality research development that led to **3** publications

AMBassadors for sustainable transITION, *AMBITION* KE & NL – 2023/24

- Change management: developed sustainable finance transition practices in schools and companies across Africa and Europe

Affiliations & Certifications

Member of the European Economic Association

Member of the AiIG - Italian Association of Management Engineering

Canadian Securities Institute, Moody's Analytics Training Certification Services

Regulatory requirements to perform securities and mutual fund transactions in many financial services positions

Trading and Portfolio Management

Asset returns, OHLC, and implied volatility. Forex market infrastructure, fixed, floating, managed (dirty) float regimes. Directional options trading strategies and usefulness filtering. Marginal benefits of options positioning

International conferences

2023

- XXXIV AiIG - Humanizing technologies for a better society
- IAFDS - International Accounting & Finance Doctoral Symposium
- ICEE - Tenth Italian Congress of Econometrics and Empirical Economics

2022

- EEA-ESEM - 37th European Economic Association and 74th European Econometric Society meetings
- XXXIII AiIG - Redesigning networks and supply chains in times of transition
- IAFDS - International Accounting & Finance Doctoral Symposium

2019

- DIGITOMICS - International Scientific Conference Economics of Digital Transformation

Academic Skills

- Faculty and student mentorship with an emphasis on team development and academic leadership
- Proficiency in statistical testing and advanced econometric methods, tailored for applied research and instruction
- Expertise in quantitative analysis with a focus on practical interpretation and academic rigor
- Curriculum design and enhancement of academic programs in finance and related disciplines
- Development of academic-industry partnerships and collaborative research opportunities
- Effective course planning, independent project management, and multitasking in dynamic academic environments
- Strategic communication for academic publishing and conference presentations
- Analytical problem-solving and academic decision-making aligned with educational objectives
- Continuous process improvement in teaching methodologies, using student feedback and evaluation
- Cross-disciplinary collaboration and active engagement with academic and industry stakeholders
- Research project management and academic publication planning, from initial proposal to dissemination

Programming languages

Python, R, Matlab, Stata, Office, LaTeX. Electronic trading platforms (Bloomberg, Interactive Brokers)

Languages

English, Italian, Spanish

Interests

aperitivo (Italy's finest networking format), tennis (4.5 USTA), soccer (Avellino), avid reader, TV shows binge-watcher (*"It's All Good Man"*), dry wit and unconventional humor